Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/09/2026
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Project Title:	Directed Task Research Study 2 Supplement Proposal: Delineate Effects of Combined Exposure to Space Radiation and Sleep Fragmentation on Brain Neurochemistry and Cognition		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02478	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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Flight Program:			
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Key Personnel Changes/Previous PI:			
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Task Description:	Deep space missions to Mars and beyond involve significant challenges for astronauts – including extended exposure to spaceflight stressors such as galactic cosmic rays (GCR), altered gravity, social isolation/confinement, and prolonged sleep disruption. These spaceflight stressors alone, or in combination, can alter neurobiological function at multiple levels in the brain to impair neurocognitive performance and jeopardize mission success. Limited information is currently available on how exposure to spaceflight stressors alone, or when combined, alters multiple neurotransmitters in key brain regions involved in complex neurocognitive function. Given that neurotransmitters in the brain provide a direct readout of central nervous system (CNS) function at multiple levels (including behavioral, emotional, stress, and cognitive function), it is critical to fully map how exposure to space-relevant spaceflight stressors alone, and when combined, alters temporal and regional neurotransmitter activity in vivo, and whether the observed neurochemical alteration in the brain impacts neurocognitive performance. Results from our prior and ongoing work on impact of individual spaceflight stressors alone has provided a solid foundation for the proposed multimodal assessment of the impact of combined sleep fragmentation and acute or chronic galactic cosmic radiation (GCR) exposure on behavioral/neurocognitive outcomes and associated neurochemical, molecular, and peripheral biomarkers in male and female subjects. Given that, with the Artemis missions, the next footprint on the Moon, and likely first one on Mars, of United States astronauts will be a woman, it will be critical to conduct studies of combined exposure in both male and female mice such that appropriate comparisons can be made between males and females, and sex differences, if any, are established. Thus, the proposed studies will continue this essential work in male and female mice and will significantly advance our understanding of how combined exposure to sleep fragmentation and acute or chronic GCR impacts CNS functionality. It is noteworthy that the Artemis mission plans to initially conduct short-duration crewed missions followed by long-duration missions with the goal of establishing a permanent base on the Moon to facilitate the feasibility of human deep space missions to Mars. Thus, astronauts on the first crewed Artemis missions to the Moon are likely to be exposed to combined spaceflight stressors of space radiation and sleep fragmentation for a shorter duration compared to longer duration missions. Consequently, using a within-subject design, we plan to assess the impact of combined exposure to 15 cGy all-at-once acute 33-beam GCRSim and 6-hr or 10-hr continuous 5 days/week sleep disturbance on the below endpoints. Next, using a within-subject design, we plan to assess the impact of combined exposure to 15 cGy given chronically 33-beam GCRSim and 6-hr or 10-hr continuous 5 days/week sleep disturbance on the below endpoints. Both the acute and chronic doses of GCR exposure are based on doses that are relevant for the lunar mission. Together, these spaceflight stressors will allow us to meaningfully compare the impact of combined spaceflight stressors on brain neurochemistry and complex behavior/cognition. This systematic approach will permit feasibility of conducting GCR exposures at BNL and allow us to meaningfully determine whether combined spaceflight stressor exposure alterations in neurotransmitter biomarkers that cause neurocognitive deficits in male and female subjects is associated with changes in molecular activity that can be linked to changes in peripheral metabolite and lipid biomarkers. Here, we will use 6-month-old adult male and female C57BL/6 mice and GCR/sleep fragmentation exposure that is identical to those utilized previously in our laboratory. In all studies, all subjects (except controls) will first be exposed to the relevant acute or chronic 33-beam GCRSim and sleep fragmentation at BNL. Next, subjects will be shipped to McLean Hospital where they will undergo a mandatory 8-week quarantine period. Thereafter, performance on the complex neurobehavioral touchscreen task will be monitored. After completion of the behavioral task, all subjects will be prepared for microdialysis studies to evaluate changes in neurotransmitter function followed by assessment of other molecular and peripheral biomarkers. We intend to use the below endpoints to study effects in male and female subjects. These include: a) assessment of 5 key neurotransmitter changes in the PFC and VTA using in vivo microdialysis analysis; b) complex neurobehavioral touchscreen (economic demand, PVT) performance. These tasks vary in difficulty and address different aspects of complex behavior and neurocognitive function; c) molecular endpoints of synaptic function and inflammatory markers (n=4-5 per group); d) peripheral plasma biomarkers, including a full metabolomic and lipidomic analysis.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	This research will provide new information not only on how combined spaceflight stressor impact neurobiology in astronauts but also humans on Earth. Studying such effects will provide a better understanding of the impact of radiation-related therapy for cancer patients on Earth as well as determining safety standards for humans on Earth. Assessing the effects of sleep disruption will improve our understanding of its impact on chronic CNS-related diseases, develop new treatment strategies for individuals suffering from sleep disorders on Earth, and for determining optimal human performance and safety standards.
Task Progress:	During this project period, evaluation of the combined exposure to sleep fragmentation and acute 33-bean GCR exposure was planned and set up for the October 2025 run at NASA Space Radiation Laboratory/Brookhaven National Laboratory (NSRL/BNL) in 36 male mice: n = 12 controls; 12 (6-hr/day/5days + acute 33 beam GCR); and 12 (10-hr/day/5days + acute 33 beam GCR). Evaluation of combined sleep fragmentation and chronic exposure to 33-beam GCR will be initiated/planned upon completion of the above studies. Specifically, the following was accomplished during this project period: • Institutional Animal Care and Use Committee (IACUC) protocol for combined exposure to sleep fragmentation and GCR approved at McLean Hospital for the October 2025 run at NSRL/BNL. • IACUC protocol for combined exposure to sleep fragmentation and GCR approved at BNL for the October 2025 run at NSRL/BNL. • Drs. Desai and Kangas completed all radiation and safety training at NSRL/BNL, as well as all necessary forms and requirements to conduct studies at NSRL/BNL. • Drs. Desai/Kangas visited NSRL/BNL facilities, including the animal facility, in early June 2025 and conducted a

	NSRL/BNL dry run with Drs. Peter Guida and Michael Sivertz for the upcoming study in October 2025. • Met with Dr. Guida, BNL support staff, the animal facility manager, and NSRL staff to plan upcoming October 2025 experiments. • Combined sleep fragmentation and acute 33-beam GCR exposure study planned starting Oct 19th – Oct 24th in male mice. Note: a similar study in female mice will be conducted in the Spring 2026 run.
Bibliography Type:	Description: (Last Updated: 01/23/2026)